1. Understand and utilise constructs based on definitions, rules and formal systems including employing familiar algorithms (Point 5 in the list)

This is the core ability required to solve this problem. Students must follow explicit mathematical rules, specifically the given formula $D = \frac{dv}{60n}$, and apply algebraic operations to rearrange this formula to solve for the unknown variable v.

- **Specific Analysis:** Students need to rewrite the original formula as $v = \frac{D \times 60n}{d}$. They must then substitute all the numerical values provided in the question ($D = 50$, $n = 3$, $d = 25$) into this new formula and perform the arithmetic calculation to find the final answer. This is a direct application of mathematical formula manipulation and numerical algorithms.

2. Use a multi-step procedure leading to a mathematical solution, conclusion or generalisation (Point 12 in the list)

Solving this problem involves several logical and computational steps.

- **Specific Analysis:**
  1. Understand the problem and identify known variables (D, n, d) and the unknown variable (v).
  2. Rearrange the formula to isolate v.
  3. Substitute the known numerical values into the rearranged formula.
  4. Perform the calculation (multiplication and division) to find the final result.
     This is a clear, linear multi-step process.

3. Manipulate numbers, graphical and statistical data and information, algebraic expressions and equations, and geometric representations (Point 7 in the list)

The task requires students to manipulate algebraic equations and specific numerical information.

- **Specific Analysis:** Students need to manipulate the formula $D = \frac{dv}{60n}$, transforming it into $v = \frac{D \times 60n}{d}$, then replacing variables with numerical values and performing arithmetic operations. This involves converting and manipulating data between symbolic representation (formula) and numerical data.

4. Articulate a solution, showing and/or summarising and presenting intermediate mathematical results (Point 8 in the list)

While the question doesn't explicitly ask to "Show your work," providing clear steps is standard practice in medical calculations to demonstrate the correctness of the answer.

- **Specific Analysis:** A complete solution should clearly show the formula rearrangement, the substitution of values, and the final calculation result (e.g., 1800 mL).

1. 利用基于定义、规则和形式系统的构建，包括运用熟悉的算法 (列表中的第 5 点)

这是解决此问题的核心能力。学生必须遵循明确的数学规则，即给定的公式 $D = \frac{dv}{60n}$，并应用代数运算来重新排列该公式，以求出未知变量 v。

- **具体分析：** 学生需要将原始公式重写为 $v = \frac{D \times 60n}{d}$。然后，他们必须将题目中提供的所有数值（$D = 50, n = 3, d = 25$）代入这个新公式中，并执行算术计算得出最终答案 (p. 1)。这是对数学公式操作和数值算法的直接应用。

2. 利用多步骤程序得出数学解、结论或概括 (列表中的第 12 点)

解决此问题涉及多个逻辑和计算步骤。

- **具体分析：**
  1. 理解问题并识别已知变量（D、n、d）和未知变量（v）(p. 1)。
  2. 重新排列公式以隔离 v。
  3. 将已知数值代入重新排列的公式。
  4. 执行计算（乘法和除法）得出最终结果。
     这是一个清晰、线性的多步骤过程。

3. 操作数字、图形和统计数据及信息、代数表达式和方程式以及几何表示 (列表中的第 7 点)

该任务要求学生操作代数方程式和具体的数字信息。

- **具体分析：** 学生需要操作公式 $D = \frac{dv}{60n}$，将其转化为 $v = \frac{D \times 60n}{d}$，然后用数值替换变量并执行算术运算。这涉及在符号表示（公式）和数值数据之间进行转换和操作。

4. 阐明解决方案，展示和/或总结并呈现中间数学结果 (列表中的第 8 点)

虽然题目没有明确要求“展示你的工作”，但为了证明答案的正确性，提供清晰的解题步骤是医疗计算中的标准做法。

- **具体分析：** 一个完整的解决方案应该清晰地展示公式的重新排列、数值代入以及最终的计算结果（例如，1800 mL）。